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Serious Game for Children with Autism to Learn Vocabulary: An Experimental Evaluation

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ABSTRACT
The use of serious games (SGs) to provide intervention for various skills to children with autism spectrum disorder (ASD) has increased in the recent years. However, the potential of these games has not been utilized to provide learning of vocabulary to children with ASD. In designing games for children with ASD, there is a need to consider components specifically from the perspectives of ASD and the generic components that can support from the theoretical, content, and game design aspects. In the absence of any existing framework to design games for children with ASD, this article first introduces a framework named serious game design framework (SGDF), constructed to provide support throughout the design process. This framework was constructed based on the extensive review of the literature on ASD and existing SGDFs. The article then focuses on the design and development of a SG prototype based on SGDF. An experimental evaluation prototype was conducted to examine the effectiveness using the withdrawal design of single-subject research design (SSRD) in improving the receptive identification of vocabulary items among children with autism before and after using the prototype. The receptive identification of vocabulary items is measured in terms of the number of correct responses given and the number of attempts made to identify the correct response. The pre- and post-evaluations of the SG prototype show that learning of vocabulary items among children with ASD improved after using the game and they retained items at the end of weeks 1 and 2 following the withdrawal of intervention.

KEYWORDS
Autism spectrum disorder (ASD); language comprehension skills; vocabulary; computer-based intervention (CBI); serious game prototype evaluation

1. Introduction

According to the fifth version of Diagnostic and Statistical Manual of Mental Disorders, referred to as DSM-5, published on May 18, 2013 (American Psychiatric Association [APA], 2013), autism spectrum disorder (ASD) is a neurodevelopmental disorder that is characterized by the impairment in social communication (verbal and nonverbal), social interaction, and restricted or repetitive patterns of behaviors. The symptoms appear during the first three years of a child's life and affect the later course of life. Every child with ASD is different; one may be very verbal, bright, and engaged, while another is nonverbal and intellectually challenged (Whalon & Hart, 2011). The research has shown that many parents take their children at a very early age for the evaluation of delay in language development (Dahlgren & Gillberg, 1989; De Giacomo & Fombonne, 1998). Eigsti, De Marchena, Schuh, and Kelley (2011) conducted a developmental review on language acquisition among individuals with ASD. The data based on the studies reviewed suggest that language impairments are present across the individuals with ASD. According to Howlin (2003), children with ASD start to produce the first word when are 38 months old, compared with 8 and 14 months in typically developing children (TD). A child with ASD does not follow a typical developmental sequence (Schreibman & Ingersoll, 2005). They also have different learning paces (Volkmar, Lord, Bailey, Schultz, & Klin, 2004) as compared to TD. There is no single standardized solution for children with ASD. However, there is a need to identify the strengths and weaknesses of each individual child with ASD and then identify an appropriate solution that may work for that individual child. Therefore, specialized syllabus and individualized support are required to meet the specific needs of an individual child (Iovannone, Dunlap, Huber, & Kincaid, 2003). The new solutions need to be tested to determine the purpose and identify target children with ASD for whom it would be effective (R. L. Simpson, 2005). There are positive and negative effects of each approach used in the research conducted for children with ASD. Marks et al. (2003) have described techniques including visual schedule, allowing children to spend more time on the tasks and use specially modified or developed material for children with ASD to stay focused on the task, repeatedly do the practice, and learn. On the contrary, Harris et al. (2015) found efficient learning in adults with ASD, followed by poor learning when the location of the target to be identified as a part of their experiment was changed (over-specificity). The authors found that over-specificity can be eliminated by reducing the stimulus repetition.

The literature on children with ASD and their learning of language has shown that there are a number of children who can effectively decode text but struggle with language comprehension, which is to understand the meaning of the text (Khowaja & Salim, 2013). Learning vocabulary directly influences language comprehension (Biemiller, 2003). Vocabulary is

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the most essential and basic component of the language; it is composed of a group of words, which provides a building block to develop any language. The vocabulary of an individual constantly evolves as they start reading different materials and communicating with others. Learning vocabulary directly influences reading comprehension (Biemiller, 2003). To properly understand the text being read, it is important to know 95% of the words in the same text (Fukkink, Hulstijn, & Simis, 2005). This indicates that processing at the word-level is considered an essential part of the comprehension. The trend of writing review papers (systematic literature review (LR) paper, survey paper, meta-analysis, etc.) by researchers to investigate a set of questions related to the area of research or particular issues has increased in the recent past.

The search on review papers related to the intervention of vocabulary among children with ASD revealed two review papers (Khowaja & Salim, 2013; Ramdoss et al., 2011). The authors of both review papers found the similar four studies related to the intervention of vocabulary (Bosseler & Massaro, 2003; Massaro & Bosseler, 2006; Moore & Calvert, 2000; Whalen et al., 2010). This shows that intervention of vocabulary among children with ASD has not been given much research attention.

CBI, which is widely used in the special education sector, has been suggested as a supporting tool for teachers of children with ASD (Hassan et al., 2011; Higgins & Boone, 1996; Powell, 1996). A number of review articles (Fletcher-Watson, 2014; Khowaja & Salim, 2013; Ramdoss et al., 2011, 2012, 2011) have been written on the use of CBI for the intervention of different skills related to children with ASD. From the reviews, a number of studies were found in which serious games (SGs) were used to provide intervention of different skills to children with ASD.

According to Abt (1987), SGs have an explicit purpose of providing an education of some skills to users and should not be played solely for the purpose of entertainment or amusement. From the LR, it is apparent that there is no single accepted definition of SGs. From one perspective, the use of the word "serious" means embedding some pedagogical elements into the games (Iuppa & Borst, 2006), whereas, from another perspective, "serious" refers to the purpose of these games (Zyda, 2005). Susi, Johannesson, and Backlund (2007) have suggested that SGs are an instantiation of game-based technology as they are used to solve problems in real life that are either too critical or too expensive; for example, education, airfield, healthcare, sales and marketing, and army. Sawyer (2007), in "Ten Myths about Serious Games," noted that learning through SGs is not as fun and entertaining as it is by using any typical games; however, SGs are developed keeping one fact in mind: their primary purpose is to provide learning to their user and fun may be a part of it. Most importantly, one can even learn without having fun and yet playing games at the same time. The literature on SGs has stressed the need to integrate the SG design and game-based learning strategies with the appropriate instructional strategies along with the pedagogical theories to provide a better learning experience to the player in the game environment (Kebritchi & Hirumi, 2008), as cited in (Rooney, 2012). From the review papers on children with ASD and SGs (Noor, Shahbodin, & Pee, 2012; Zakari, Ma, & Simmons, 2014), it is evident that various SGs have been developed to support these children in the learning of different skills; however, they have not been used to provide learning of vocabulary to children with ASD.

Considering the fewer number of studies on vocabulary learning of children with ASD, despite its importance, and underutilization of SGs to support these children in the learning of vocabulary, this research attempts to investigate the development of SG and its experimental evaluation among children with ASD to learn and improve their vocabulary.

2. Serious game design framework (SGDF)

2.1. Motivation

The trend to construct specialized SGDFs has increased in the recent past (Arachchilage & Love, 2013; Denis & Jouvelot, 2005; Ijsselsteijn, Nap, De Kort, & Poels, 2007; Järvinen, 2009; Moreno-Ger, Burgos, Martínez-Ortiz, Sierra, & Fernández-Manjón, 2008; Shoukry, Sturm, & Galal-Edeen, 2012). The components in the specialized design frameworks are tailored to suit the need of the users (children with ASD for vocabulary learning in this research) and the design requirements of the SGs, which are otherwise not possible through the generalized framework. However, this trend has not been seen in the context of children with ASD as only two frameworks were found (Park, Abirached, & Zhang, 2012; Whyte, Smyth, & Scherf, 2015). The framework by Park and colleagues only considers components from the pedagogical perspective, which provides guidance on how emotions can be taught step-by-step to children with ASD. The other main components, especially from the perspective of autistic children and the components that facilitate in the design of an SG in general, are not considered. Whyte et al. (2015) have focused and identified seven principles for the design of SGs for the motivation and engagement of individuals with ASD in the game environment, as skills to be learnt may be difficult and may require a huge amount of time (weeks or months). There are many other aspects of design principles that are important to consider in the game environment (Khowaja, 2017); these include positive feelings, surprise, fantasy, and mystery among others. Therefore, both frameworks can be regarded as meeting the minimal requirements in terms of components to design games for children with ASD to learn vocabulary.

In the unavailability of existing frameworks to design SGs that can facilitate children with ASD in the learning of vocabulary, a framework is constructed to support in the design process of the SG. The design of SG can then be transformed into a prototype and evaluated with children with ASD. The following subsections provide details on the construction and evaluation of SGDF.

2.2. Construction of SGDF

Existing theory and prior research, which is recommended as one of the methods by Maxwell (2012), were used as a basis to construct a framework to design SGs for children with ASD. A three-step process was carried out to construct SGDF:

(1) identify components that need to be incorporated into the framework,
(2) identify underlying structures that can be used as a basis to construct a new framework, and

(3) logical placement of these components in the identified structure.

The first two processes are the outcomes of the review of the literature, while the third process uses each component and its purpose from the first phase and maps it to the identified structure from the second process. Each of these processes is briefly described in the following subsections.

2.3. Identification of components

Figure 1 shows the LR sources that contributed to the identification of components for the framework based on the two systematic and five narrative reviews. The sources, namely strategies and CBIs, modalities, autism behaviors, instruction methods, SGs, and common components in SGs for ASD, were found from the literature on children with ASD and their learning of vocabulary. The SLRs are represented by the dash rounded rectangle, whereas the NLRs are represented by a solid rounded rectangle. An arrow from one source to another shows how the review of current source uncovered the more sources to be reviewed. Table 1 shows the classification of components found from the literature reviewed into three perspectives: children with ASD, typical children, and typical users based.

2.4. Identification of structure

The structure based on the input, process, output (IPO) model proposed by Garris, Ahlers, & Driskell (2002) was selected for the development of the framework as 1) it is widely used to construct other frameworks, 2) it is used to develop games, and 3) a similar process as of IPO is also used in the studies related to ASD.

2.5. Logical placement of components in structure

Each identified component needs to be logically placed in the phases of the IPO model; this is to ensure that both the structure of framework and the components are synchronized with that of IPO by Garris et al. (2002).

2.6. Evaluation of SGDF

The initial version of the proposed framework was thoroughly evaluated through multiple methods of evaluation, which include expert review studies, and an applicability test. The methods used and the outcome produced throughout the evaluation are shown in Figure 2. The expert review studies were conducted to analyze the components used, their usefulness in the framework, relations between components, and the structure of the proposed framework. An applicability test was performed to analyze the application of the framework to design an SG for the vocabulary acquisition of children with ASD. The proposed version of the framework was produced based on the comments provided by the experts across the above-mentioned studies, test, and survey.

The proposed version of the SGDF is shown in Figure 3. The descriptions of all the components and the details of all the processes carried out in the construction of SGDF are available in Khowaja & Salim (2017). The descriptions of all the components are also appended in Appendix A. During the applicability test, experts were asked to produce game design document using the framework proposed in this research. A design produced by one of the experts was transformed into an SG prototype; the details of the design are appended in Appendix B. The development and evaluation of the prototype are discussed in sections 3, 4, and 5.

Table 1. Components contributed from different perspectives.

Perspective	Components
Learning of vocabulary among children with ASD	Autism behavior and symptoms, instruction method, instructional strategies, modalities
Typical children	Learning activity, game story, narratives, non-player characters, reflection, user profile
Typical user	Desirable capabilities, instructional contents, learning outcomes, game genre, game mechanics, game dynamics, storytelling, game-based learning attributes, user achievements, debriefing, context

3. Game development

3.1. SG prototype

As mentioned in the earlier section, a design produced by one of the experts from the applicability was used and it was transformed into an SG prototype. The design of an expert was used to demonstrate that experts were able to transform the concept of the game using SGDF into the game design document, which was then transformed into the prototype by the authors of this research article. The user interface guidelines for children with ASD by some authors (Davis, 2008;

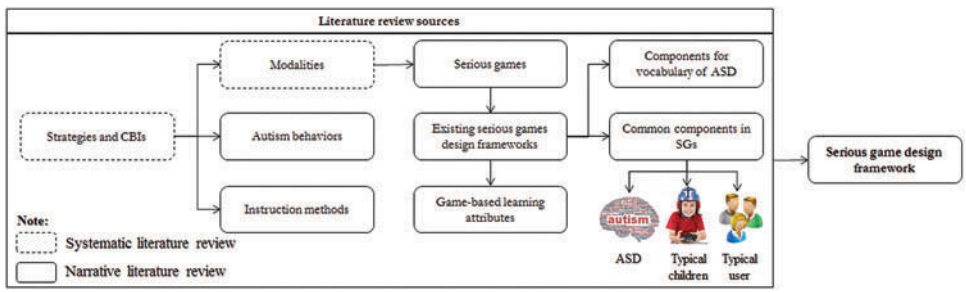

Figure 1. Literature sources used for the identification of components.

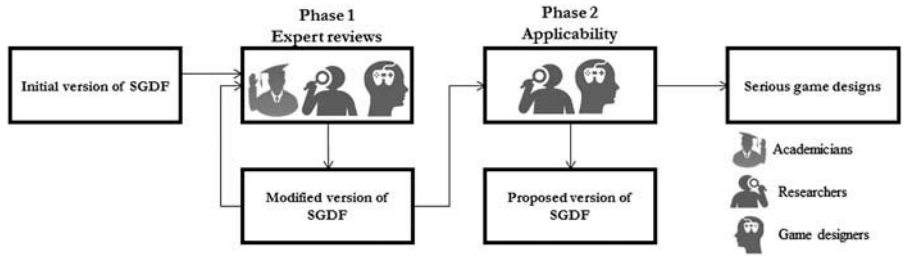

Figure 2. Methods and flow of evaluations of the proposed framework.

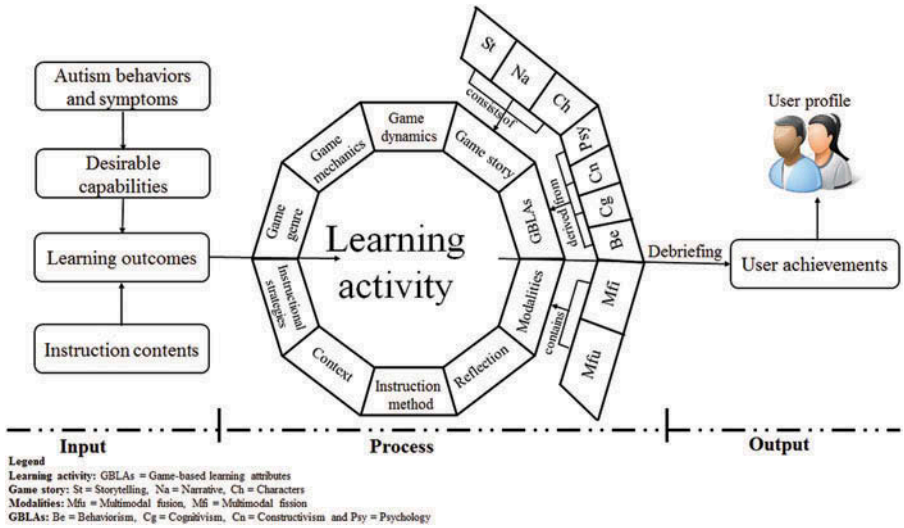

Figure 3. Proposed version of the serious game design framework (Khowaja & Salim, 2017).

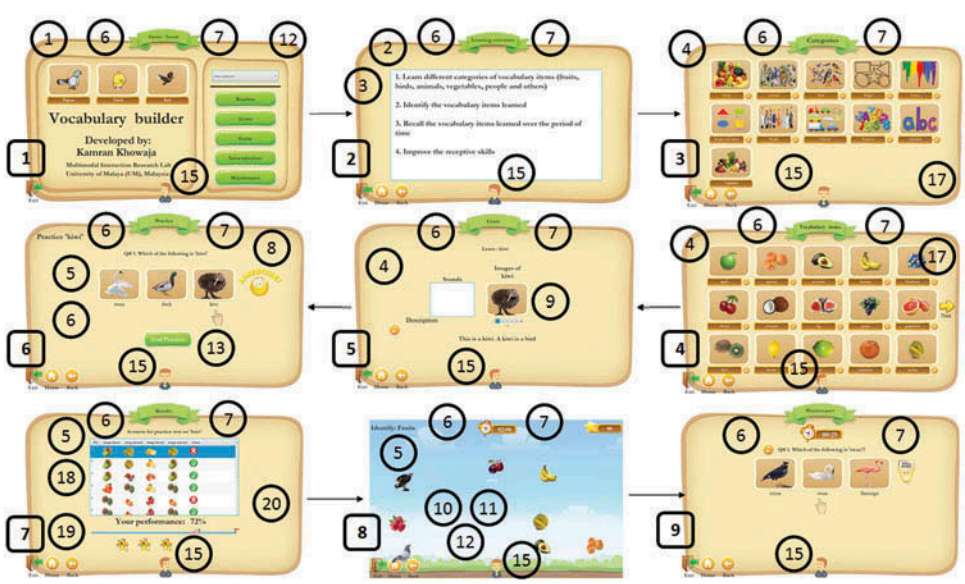

Figure 4. Mapping of components from scenario to game.

Higgins & Boone, 1996; Learning Styles of Autistic Children, n.d.) were considered while developing the SG prototype.

Figure 4 shows different screens of the prototype; the sequence of screens shows the flow followed in typical SGs. Each screen has used one or more components from the SGDF; the number in each circle corresponds to one of the components presented in Table 2, while the number in each rounded rectangle represents the screen number. The arrows between screens show that the user can move from the current screen represented by the tail of an arrow to the next screen represented by the head of an arrow.

Table 2. Components of serious game design framework proposed in this research.

S#	Component	S#	Component
1	Autism behaviors and symptoms	11	Game dynamics
2	Desirable capabilities	12	Game genre
3	Learning outcomes	13	Reflection
4	Instruction contents	14	Game story
5	Learning activities	15	Characters
6	Game-based learning attributes	16	Narratives
7	Modalities	17	Storytelling
8	Instruction method	18	Debriefing
9	Strategies	19	User achievements
10	Game mechanics	20	User profile

The overall flow of the game across the screens of Figure 4 is shown in Figure 5; the rectangle represents the screen while the direction of an arrow indicates that the player (child with ASD playing game) can move from one screen to another. For instance, the two-ended arrow between the "Start" and "Play activity game" screen indicates that the player can move between both of the screens, while an arrow from "Activity game" to "Start" shows that the player can move from "Activity game" screen to "Start" but not vice versa. The description of each screen is provided in subsection 3.2.

As discussed in section 1 that components of SGs are typically supported by pedagogical theories, this is also the case with the prototype shown in Figure 4. Table B3 presents a total of 16 game-based learning attributes (GBLAs) identified by the expert that have been incorporated across the different screens of the prototype based on their description and usage. For instance, the help and support based on the theory of constructivism are provided throughout the game; the rules followed across the game based on the theory of behaviorism are described in Table B2. These attributes have been chosen from the set of 33 attributes identified by Khowaja (2017). Each attribute represents the successful practices from a typical classroom environment so that learning of player is validated in the game environment using a similar way as of a typical classroom. The GBLAs are based on four theories, including behaviorism, cognitivism, constructivism, and the theory of psychology; the first three theories belong to the theories of learning. A brief description about these theories is presented below, while details on these theories and the classification of attributes to theories are presented in Khowaja (2017).

(1) **Behaviorism**: The theory of behaviorism is about the study of behaviors that can be observed and measured (Good & Brophy, 1990). According to this theory, learning is organized as a series of activities. The player of the game (child with ASD) can take part in various activities throughout the game. The sequence of responses provided by the player needs to be recorded. Therefore, practicing can allow the player to achieve the desired results.
(2) **Cognitivism**: Cognitive theory explains behaviors of humans by understanding how learning is processed inside the brain. The theory deals with explaining human behaviors by understanding how learning is processed inside the human brain. The cognitive load of the player needs to be reduced to ensure that the player is able to process information easily; this will improve their performance and increase learning.
(3) **Constructivism**: According to constructivist theory, learner uses their own experience to create and organize information. The theory is based on the principle that learning involves active processes. The player takes part in different learning activities of the game. From each learning activity, they may encounter new information; they use this information and their experience to construct new knowledge and schemas.
(4) **Theory of Psychology**: While the above three theories highlight all about effective learning of the player, it is also important to understand how a player thinks and behaves during the learning. Therefore, by understanding the answers to two types of questions (i.e., the how's and the why's of human thoughts and behaviors) better, the theory of psychology seeks to understand more about the players (learning (Poulou, 2005). This can be simplified into positive and negative emotions. Eagerness, attention, engagement, pleasure, perseverance, trying hard, and attempting a test, are some of the major

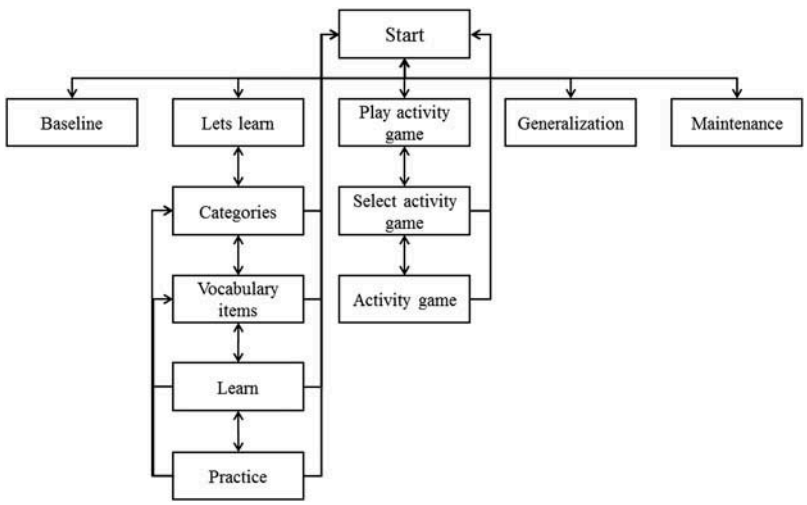

Figure 5. Flow of game between screens.

causes for positive emotions (Malone, 1980; Preston & Morrison, 2009). However, nervousness, feeling unprotected, fear of contest, fear of failure, and inferiority are some of the leading triggers of negative emotions.

4. User interface of the game

This subsection provides brief details of all the screens of the game shown in Figure 4.

4.1. Screen 1 – main screen

This screen gives access to five different options (baseline, generalization, maintenance, learn, and game) available in the game that the player can select with the help of a caregiver, teacher, or any other person sitting along with him or her. Baseline, generalization, and maintenance are three different types of tests; the details of these tests are provided in the next section. The learn option allows the player to study different vocabulary items divided into various categories, while the game option allows the player to play a game in order to determine the impact of learning different vocabulary items.

4.2. Screen 2 – learning outcomes

This screen displays the learning outcomes of playing this SG to the player; this way the player is aware of the anticipated outcome of using the game.

4.3. Screen 3 – categories

This screen shows all the categories of vocabulary items that can be learned through the game. The rectangle with round dots surrounding a category, for instance "birds," shows that there are vocabulary items within this category that the player can learn. However, the rectangle with dashes surrounding a category for instance, "internal body parts," shows that it is currently disabled as there are no items available within this category for him or her to learn. To facilitate the player in the selection of a category, an image of the category is shown as well as the name of the category underneath the image; moving the mouse cursor in the region of any category will read aloud the name of that category to the player. Furthermore, the player can also hear the name of any category by clicking on the music icon beside the name. If the number of categories available in the game is more than 15, then they are split into multiple screens where each screen contains 15 categories and the player can move between screens by clicking on the previous or next buttons. The prototype currently supports the learning of 11 categories of vocabulary items, which means that all categories are presented on a single screen. These categories include fruits, vegetables, animals, birds, shapes, colors, colors and shapes, people, vehicles, numbers, and alphabets.

4.4. Screen 4 – vocabulary items

When a player clicks on any of the categories, the next screen is shown to the player, which displays all the vocabulary items present in the selected category from which the player can select any item to learn. If the number of vocabulary items in any category is more than 15, then they are split into multiple screens where each screen contains 15 vocabulary items and the player can move between screens by clicking on the previous or next buttons. To facilitate the player in the selection of a vocabulary item, an image of vocabulary is shown as well as the name of the item underneath the image and moving the mouse cursor in the region of any item reads aloud the name of that item for the player. When he or she clicks on any of the vocabulary item, the next screen of learning is shown, which is described in the next subsection. The game currently supports 209 vocabulary items altogether, which is divided into 11 available categories, as presented in Table 3.

4.5. Screen 5 – learn

The learn screen of a vocabulary item displays the following. 1) All the images associated with an item in a paginated view so that the player can browse through the images one after another by clicking on either the index number of an image or the previous or next buttons. 2) All the available pronunciations of an item in the list box of sounds; each pronunciation can be played by a single click on the name over and over again to improve their receptive skills. These pronunciations are downloaded from the website of Dictionary.[1] 3) It also displays a brief description of a

Table 3. Vocabulary items supported by game.

Category	Vocabulary items
Fruits	Apple, grape, apricot, grapefruit, avocado, kiwi, banana, lemon, blueberry, plum, lime, cherry, mandarin, coconut, melon, fig, orange, papaya, pomelo, peach, quince, pear, pineapple, strawberry, pitahaya, watermelon, pomegranate
Vegetables	Carrots, potato, onion, spinach, broccoli, cucumber, eggplant, okra, peas, chickpeas, coriander, mint, kidney beans, corn, ginger, garlic, cabbage, cauliflower, zucchini, sweet potato, chili, bell pepper, bitter gourd, beetroot
Animals	Dog, crocodile, cow, mouse, cat, snake, giraffe, horse, donkey, buffalo, sheep, pig, panda, rabbit, dinosaur, camel, lion, elephant, tiger, zebra, monkey, deer, chimpanzee, goat, rhino, bear, hippopotamus
Birds	Pigeon, swan, chicken, crow, duck, owl, parrot, turkey, flamingo, ostrich, peacock, kiwi, penguin, myna, sparrow, eagle, woodpeckers, kingfishers, vultures, seagull
Shapes	Square, triangle, right triangle, pentagon, circle, rectangle, line, rounded rectangle, parallelogram, trapezoid, diamond, hexagon, heptagon, octagon, decagon, dodecagon, pie
Colors	Blue, pink, green, white, red, black, grey, orange, yellow
Colors and shapes	Blue circle, red circle, blue square, red square, blue triangle, red triangle
People	Doctor, fireman, postman, chef, farmer, police officer, barber, carpenter, plumber, electrician, singer, painter, engineer, dancer, drummer, keyboard artist, pianist, guitarist, nurse, judge, teacher, sweeper
Vehicles	Car, truck, fire truck, bus, school bus, tractor, helicopter, aeroplane, bicycle, scooter, motorbike, tank, train, ambulance, cable car, crane, ship, forklift truck, submarine, road roller, bulldozer
Numbers	0, 1, 2, 3, 4, 5, 6, 7, 8, 9
Alphabets	A to Z (uppercase and lowercase)

[1] http://dictionary.reference.com/.

selected item that can be read aloud for a player when they click on the music icon. 4) Lastly, the practice button is only shown on the screen when the player has browsed through all the images at least once; this is to ensure that the player has seen all the images of the item and know how the item may look like before he or she attempts a practice test. The number of images shown for each vocabulary item varies from item to item. For instance, an apple can be in different color based on the brand, but the view of an apple, for instance (front, back, left or side, top, or bottom among different angles), is almost the same; the interior view of an apple is also shown so that they can learn how it looks like from the inside. Similarly, a vocabulary item in the category of animals or birds among others can be sitting, running, or flying in different directions; therefore, images of each item are shown from different views so that the player can observe and memorize each item well. The number of images shown for each vocabulary item can allow the player to generalize their learning about the item. However, this generalization within the natural environment is not tested as a part of this research.

4.6. Screen 6 – practice

The purpose of practice is to examine whether the player has retained an item just learned. Figure 6 shows how the instruction method discrete trial training (DTT) is used step-by-step to facilitate the player in the learning of a vocabulary item. Each circle has a number from one to five; these numbers correspond to the steps of DTT.

Following are the details of how five steps of DTT are used in the context of the current game.

(1) The player is informed about the practice related to the vocabulary item he or she had just learned. The player is shown a question and asked to identify the correct image of an item; altogether three images are shown, of which one is the correct image, while the remaining two images act as distractors, i.e., items that also belong to the same vocabulary category as of the correct item. For instance, if an image of an apple is shown, then the next two images shown (acting as distractors) are of different fruits. As mentioned earlier, each item has at least three images; therefore, for this practice, a random image of all three items is shown to the player. The location of the correct answer is also randomly chosen so that the player does not know where the correct answer could be placed. The names of all the items are also shown underneath the respective items.

(2) If a player does not answer a question in 5 s of time, a verbal reminder is given to the player by repeating the question again. This is repeated every 5 s until an image is selected by the player.

(3) The player selects an image of a possible correct answer and just like the practice button of Figure 6 appeared; a button of "End practice" is only made visible on the screen once the player has correctly identified three images of an item.

(4) A positive feedback in the form of verbal response and image as shown in screen 6 of Figure 4 is shown to the player as an encouragement in either case whether the player selects a correct image or an incorrect image. However, if a player selects an incorrect image, a visual hint is given in the form of an index finger pointing to the correct image for 1 s. For each attempt made to identify a correct image, information about the attempt number and the time the player took to select an image is recorded for further analysis. The attempt number is initialized to zero each time a new question is shown; the text of all questions remains the same as the question is about the same vocabulary item, but the question number is in increment by 1. Similarly, the attempt number is also increased by 1 each time the player selects an incorrect image.

(5) A gap of 2 s is given before the next question is shown to the player.

Figure 6. Step-by-step use of the instruction method DTT to facilitate learning.

4.7. Screen 7 – results

This screen presents the detailed results and the overall achievements. 1) The details about all the attempts made to answer the question and its results are presented in the form of a table; the information displayed includes: all three images that were shown to the player for each attempt, the image selected by the player, and an icon in the form of a tick or a cross to indicate whether the answer selected was correct or wrong. 2) The percentage of all of the questions correctly answered in the first attempt. Lastly, the player is given stars based on the percentage secured; five stars are given for percentage of 90 or above, four stars for percentage of 80 or above, three stars for percentage of 70 or above, two stars for percentage of 60 or above, and one star for percentage of 50 or above, while no star is shown if the percentage is below 50.

4.8. Screen 8 – activity game

Once the player has completed the lessons of all the vocabulary items in the selected category, the player is given an access to play an activity game. The practice test during learning focuses on the learning of each vocabulary item,

while the activity game focuses on the learning of all the vocabulary items within the selected category. The purpose is to determine the extent to which a player can correctly identify the vocabulary items from the selected category in the presence of distractors.

The screenshot of the game is shown on screen 8 of Figure 4; the purpose of this game is to let the player select all the items that belong to the category they learned. These items are mixed with the distractor items that belong to other categories. The activity games can be individually configured for each player; the information that can be configured in the game include: 1) test items: categories from which items need to be selected by the player; 2) distractor items: categories from which items will act as distractors; 3) total number of items (test items and distractor items) to be shown on screen at any time; this number is specified in terms of multiples of four so that distribution of test items and distractor items can be easily done; 4) background image of the game and its opacity; 5) background music of the game; 6) animated items that can move around the screen; and lastly 7) type of game (easy, medium, and hard).

Although the learning of contents through SG presented in this research is focused on the fruits only and its testing is based on two activity games, the SGs provide one facility to the caregivers and teachers of children with ASD. This facility allows them to use their imagination and create a customized activity based on the contents to be tested in the presence of selected distractors, duration and type of an activity, the total number of items (testable and distractors), background image, music, and moveable objects to be incorporated into the activity, as shown in Figure 7. They can also modify an existing activity and use it for the different instructional contents with just a few clicks.

(1) Allows selection all those categories from which vocabulary items will be treated as test items; this means that during activity game, a player will be asked to identify only those vocabulary items that belong to these lists.
(2) Allows selection of all those categories from which vocabulary items will be treated as distractor items; this means that during activity game, these items will be mixed with the items to be tested and the player needs to avoid selection of these distractors.
(3) Allows selection of total time of an activity (minutes and seconds) that a player can play. It also allows selection of the type of activity game (easy, medium, or hard).
(4) Allows selection of a total number of vocabulary items that will be shown to the player at any point in time. This includes both the testable and distractor items. The percentage of testable items against distractor items may vary.
(5) Allows percentage of test items from the total number of vocabulary items to be shown in the activity game. The activity games played by the player as a part of this research include 75% (i.e., three out of four vocabulary items) test items that act as test items.
(6) Allows percentage of distractor items from the total number of vocabulary items to be shown in the activity game. The activity games played by the children with ASD as a part of this research include 25% (i.e., one out of four vocabulary items) items that act as distractor items.
(7) Allows selection of the background image of an activity game.
(8) Allows selection of movable characters to be present in the activity game. These characters can move from the start point (x1, y1) to the end point (x2, y2) in the specified times in seconds. Once they reach the end point, they can move in the backward direction if the option of reverse is selected; else, they will again start moving from the start point.
(9) Allows selection of the background music that will be played during activity game. Children with ASD can get distracted by loud music; therefore, the volume of background music is set to 50%.

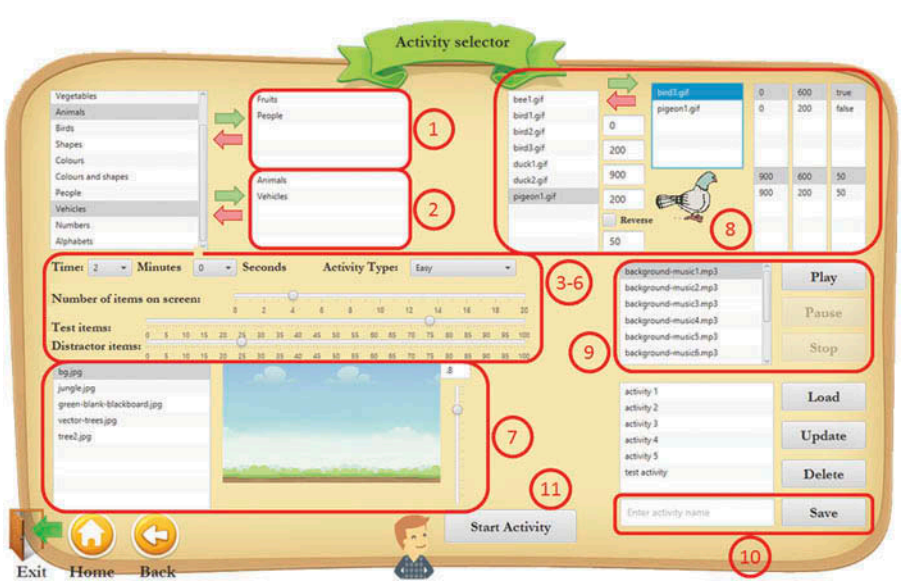

Figure 7. Activity selector screen.

(10) Allows the person to save all the configurations of the activity game in the file with an easy-to-remember activity name. This also allows a person to load configuration of an existing activity game so that it can be loaded for the player to play. The configuration of each activity game can also be edited, updated, or deleted.
(11) Creates a game activity environment by loading all of the configurations present on the screen into the environment and start activity.

The two different types of activity games created using the activity selector screen are shown in Figure 8.

4.9. Screen 9 – pretest and posttest

This screen is used to measure the performance of learning vocabulary before (pretest) and after (posttest) using prototype. The screen of the baseline is used for pretest, while the screens of generalization and maintenance are used for the posttests. The screens of both pretest and posttest are similar to those of practice with one difference: these screens also show a clock that displays the time the player has spent on the test. Similar to practice, the text of the question is shown as well as read aloud for the player, and the same positive feedback from the practice screen is given to the player regardless of the correct or wrong answer in the tests of baseline, generalization, and maintenance. However, the hint is only given in the baseline and it is provided in the same form as in the practice screen.

5. Usability study of the SG prototype

The aim of this usability study was to detect and fix as many usability problems as possible in the SG prototype before children with ASD start using the prototype to learn vocabulary. Such usability problems can distract the user while they are using the prototype; therefore, it is necessary to fix usability problems as early as possible. According to Dix, Finlay, Abowd, and Beale (2004), the evaluation of any system (SG prototype in this research) can be performed through expert analysis or user participation, i.e., potential users of the system. In this research, heuristic evaluation is used as an approach to perform an evaluation of the prototype for the following reasons (Dix et al., 2004; Rogers, Sharp, & Preece, 2011; Shneiderman, & Plaisant, 2010):

- Relatively inexpensive and fast
- Performed at any phase of product development
- Identifies many usability problems
- Achieves substantially better performance by aggregating the evaluation from several evaluators
- Provides an overview of the complete design
- Pays direct attention to particular aspects of a design and associated problems
- Does not attempt to trace specific user behavior; rather it critiques the attribute of an interface itself.

Nielsen and Molich (1990) developed a set of 10 heuristics that can be used to identify usability problems in the system. This set of heuristics is a generic set and may not identify problems in specialized systems that have been developed, for instance, healthcare system, ambient display, SGs, etc. Hence, the latest trend is to come up with more advanced heuristics for the state-of-the-art technologies and systems. Usually, researchers establish their own set of heuristics by adapting Nielsen's heuristics along with design guidelines, market study, requirement documents of a specific product, expert evaluations, and researchers' expertise in the area of research; alternatively, it may consist of a combination of these items (Rogers et al., 2011). Khowaja, Salim, and Asemi (2015) have developed a modified set of 15 heuristics to evaluate an interactive system developed for children with ASD; this modified set was used to perform usability evaluation of the SG prototype.

6. Experts and recruitment

The experts selected for this study include academic staff teaching HCI, doing research in the same area, and have experience of working with heuristic evaluation. The search for these experts was performed in Google search engine.[2] Based on the search results, seven experts were identified for this study. The email invitations were sent to these experts and three of them positively replied to the email and confirmed their participation in this study. The demographic information of these experts is shown in Table 4.

Figure 8. Activity games created using the activity selector screen.

[2]https://www.google.com.

Table 4. Demographic information of experts.

Expert#	Gender	Years of experience	Expertise
1	Female	10	Researcher
2	Female	15	Academic staff
3	Male	12	Researcher

7. Instruments used

Two instruments were used in this study. The first instrument was the SG prototype developed; the details on the prototype are presented in section 3.2. The second instrument was the modified set of 15 heuristics developed by Khowaja et al. (2015) to evaluate an interactive system developed for children with ASD.

8. Study protocol

The study procedure involved the steps listed below:

(1) Email invitations were sent to the experts. The respondents were assured that all evaluation data provided for the research would be anonymous. They were asked to provide a response to the email on whether they consented to participate in the evaluation process.
(2) Experts carried out an evaluation at their own site. A briefing session was held via Skype to inform the experts of a few aspects of the study: i) purpose of evaluation; ii) URL of the SG prototype to be evaluated; iii) modified set of heuristics to be utilised; iv) prototype exploration; v) two-phase evaluation; vi) determination of usability issues; and vii) email submission of data. The evaluation process required experts to provide a summary of the heuristic-related issues. They were also asked to rate severity of the issues on a scale of 0 to 4, where 0 represented 'not a problem'; 1, 'cosmetic problem' and 4, 'usability catastrophe'.

9. Procedure

The usability evaluation of the prototype was conducted in two phases, which are briefly described in the following subsections.

9.1. Phase 1

Experts evaluated the SG for children with ASD and submitted a report of usability problems identified through email. All the reports from the experts were compiled and listed before they were communicated to the developers of the SG. The developers worked on all the problems. All the problems were resolved by the developers, who made changes to the SG accordingly.

9.2. Phase 2

The same URL mentioned contained the latest version of the system; this enabled the experts to download the old and new copies of the systems if they wished to make comparison. The experts were emailed a copy of the report detailing the issues identified by them; it was in the form of questionnaire. The experts were requested to provide the following responses for each issue: 1) the identified issue had been resolved; 2) the identified issue was still unresolved; or 3) they were unsure of the status. The experts responded to the questionnaire via email once they had completed their evaluation.

10. Results

10.1. Phase 1

In phase one, a total of 43 problems were identified by the experts. Figure 9 shows the number of problems found in this phase, and the average severity of all the problems occurring in the SG prototype using a modified set of heuristics. For each heuristic, the stacked column shows the number of problems found in four severity ratings (cosmetic problem, minor usability problem, major usability problem, and usability catastrophe), whereas the line connecting the markers shows the average severity ratings of all the problems found. The cosmetic problems are represented with the light downward diagonal pattern, minor usability problems are represented with the light upward diagonal pattern, major usability problems are represented with the light vertical pattern, and usability catastrophes are represented with the light horizontal pattern. A The shorter name of heuristics from Table 5- is represented by a single-word name in every column. Examples of the frequently violated heuristics are match, consistency, control, and minimalist. Among all the heuristics, match heuristic was most frequently violated. The average severity rating of all the problems found is 1.6, which means that most of the problems are either minor or major.

From the analysis of the usability problems, it was found that the problems identified for the "vocabulary items" screen are

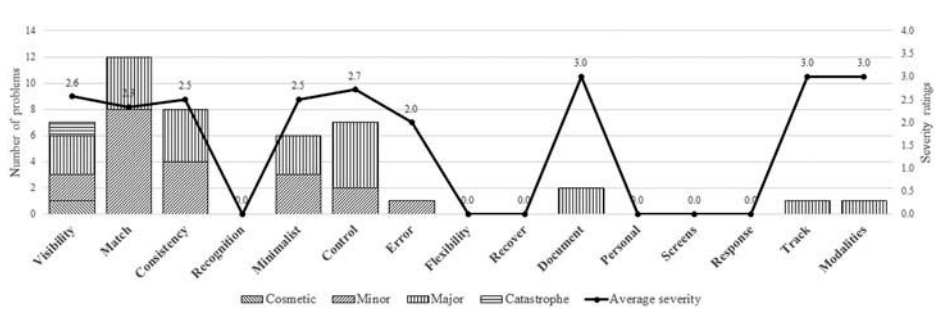

Figure 9. Number of usability problems and average severity ratings found in the serious game prototype using a modified set of heuristics.

slightly higher than the problems identified in the other screens. Therefore, the remaining part of this subsection is focused on the usability problems identified and rectification of these problems in the design of the "vocabulary items" screen. Table 5 presents the heuristics by their shorter name, severity rating, and usability problems found using a modified set of heuristics for the "vocabulary items" screen. The first five usability problems in the table are specific to the "vocabulary items" screen, while the remaining three problems are general and need to be resolved across the screens.

Figure 10 shows the two screens of "vocabulary items;" the image on the left side shows the screen before it was evaluated by the experts, while the image on the right side shows the same screen once necessary changes (highlighted by the dash rounded rectangle) were incorporated into based on the usability problems identified by the experts. The screens of the SG prototype shown in Figure 4 are the updated screens after all the changes were incorporated into based on the usability problems identified by the experts of the study.

Table 5. Examples of problems found using the modified set of heuristics for the vocabulary items screen.

Heuristic	Severity	Problem found
Control	3	There should be a button beside the description of vocabulary item that reads aloud the description of the specific item when clicked
Consistency	2	The images in the vocabulary items are too small for players to see. The size needs to be consistent with the size of images shown for the category.
Consistency	3	Just like the categories screen, place a sound button next to the name of the vocabulary item so that the player can hear the pronunciation of the vocabulary item
Consistency	2	The pronunciation of the vocabulary item should also be playable when the player moves the cursor over anywhere in the icon
Match	3	Place a label below the previous and next buttons. The previous button and its associated label underneath the button should only be shown when there is a possibility to view previous categories or vocabulary items. The same pattern should also be followed for the next button as well.
Match	3	Exit, home, and previous buttons should have labels underneath them
Minimalist	3	The sounds on exit, home, and previous buttons are lengthy; therefore, if the cursor is quickly moved in between these buttons, multiple sounds are heard simultaneously. Therefore, they should have brief sounds associated with each button
Visibility	3	The sounds on exit, home, and previous buttons are too long; therefore, if the cursor is quickly moved in between these buttons, multiple sounds are heard simultaneously. Therefore, stop the current sound being played before playing a new sound

10.2. Phase 2

In phase two, each expert was communicated via email to review the updated version of the SG prototype along with their actual comments in the form of a questionnaire. For each comment, they were asked to specify whether their identified problems have been fixed, whether their identified problems have remained unresolved, or whether they are unsure about it; they were also asked to specify if they find any new issues in the prototype. Based on the completed questionnaires received from the experts, it was found that 40 out of 43 problems were resolved, while the remaining problems remained unresolved, and one expert identified two problems in the prototype. The following three unresolved problems and the newly identified problems by the experts were also resolved in the prototype.

(1) **Unresolved issue**: When I move the mouse over different buttons, the sounds simultaneously play and make it incomprehensible. Therefore, have a smaller sound clip associated with each button.
(2) **Unresolved issue**: Categories should be grouped and presented according to the number of items available in each category. Those categories that have one or more vocabulary items to be learned should be presented first, followed by those categories that do not have any vocabulary items in them.
(3) **Unresolved issue**: On the game screen, when a player selects any object, it should show the name of the object for a moment before it disappears. The name should also be pronounced irrespective of whether the player has selected a correct or an incorrect image.
(4) **Newly identified problem**: Unable to read the buttons' caption because the Avatar comes in front of the buttons (Game Selector).
(5) **Newly identified problem**: The Avatar constantly stays visible at the same location. There should be a way to hide and unhide its appearance or reduce and restore its size.

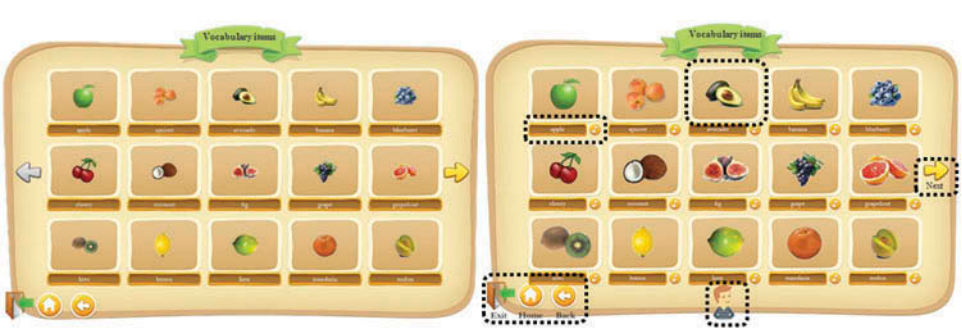

Figure 10. Screens of vocabulary items to learn.

11. Discussion

The analysis shows that from the usability problems found through the modified set of heuristics in phase one, most of the problems were fixed in phase two. The unresolved issues and new problems identified were fixed at the end of phase two. The analysis of the frequently violated heuristics from Figure 9 shows that Match heuristic was more frequently violated than the other 14 heuristics.

11.1. Experimental evaluation of the prototype

The aim of this experimental evaluation is to examine the effectiveness of the SG prototype entitled "vocab builder" in improving the performance of learning vocabulary among children with autism. The research questions guiding this evaluation are as follows:

RQ 1. Which research design is appropriate to evaluate a prototype of the game with the children?

RQ 2. How can the learning of vocabulary be measured in children with ASD?

RQ 3. Does the learning environment of a game have an impact on the acquisition and remembrance of the vocabulary of children with ASD?

11.2. Review of design for experimental evaluation

The review of the literature on children with ASD, which include systematic LRs (Khowaja & Salim, 2013; Ramdoss et al., 2011), has shown that single-subject research design (SSRD) and group-based designs are commonly used to investigate the effectiveness of intervention among children with ASD. Both of these designs are also commonly used designs across different fields including education, psychology, and health, among others (Creswell, 2012). The national research council of United States of America (USA) has highlighted that researchers working with children with ASD have frequently used the SSRD to provide intervention of some skill (National Research Council 2001). The SSRD is chosen for this study to analyze the performance of learning vocabulary among children with ASD through the SG based on various factors including applicability, measurement, data analysis, and statistics among others. The remaining part of this subsection highlights the importance and details of SSRD.

The purpose of using the SSRD design is to learn about the change in the specific behavior of an individual over a period of time. SSRD is used to show or establish a causal relationship between the intervention used in the research and the result of an individual (Neuman & McCormick, 1995). There are five different designs of SSRD (Byiers, Reichle, & Symons, 2012): 1) pre-experimental (or AB or A/B) design; 2) reversal or withdrawal (or ABA or A/B/A, ABAB, or A/B/A/B) design; 3) multiple-baseline/multiple-probe design, 4) alternating interventions and adapted alternating interventions designs, and 5) multiple-interventions design.

Table 6 presents four studies on the language comprehension skills of children with ASD, which are based on SSRD. These studies have been extracted from the reviews (Khowaja & Salim, 2013; Ramdoss et al., 2011) that have included studies related to the learning of vocabulary among children with ASD. For each study, the details presented include: 1) skills targeted, 2) material used, 3) design used, 4) data collected, 5) duration of the study, 6) data collection instances, and 7) type of analysis performed. This would support in the identification of appropriate details that can be used in the experimental evaluation of prototype with children with ASD. The remaining part of this subsection presents the analysis from each column of Table 6.

Table 6. Studies on language comprehension skills of children with ASD.

Source	Skills targeted	Materials	Design	Data collected	Duration of the study	Data collection instances	Analysis
Massaro and Bosseler (2006)	Receptive identification of pictures and vocabulary	24 (12% of at least 196 items) vocabulary items divided into 4 equal groups	Within-subject design followed alternating treatment	Number of correct receptive responses	30-min sessions, on 3 days per week until all 24 vocabulary items are correctly identified in the posttests across two consecutive training sessions	Pre-training and post-training	Visual
Bosseler and Massaro (2003)	Receptive identification of pictures and vocabulary	18 (9% of at least 196 items) vocabulary items uniquely selected for each participant and divided into 3 equal groups	Within-subject design followed single-subject multiple-baseline design	Number of correct responses	33 days	Pre-training, training, post-training and generalization	Visual
Hetzroni and Shalem (2005)	Matching word to sample	8 food items were selected for each participant. Same items are used for all participants	Multiple-probe design across participants	Number of correct responses	Not specified	Baseline, intervention, generalization, and maintenance	Visual
Coleman-Martin et al. (2005)	Promote word identification using the Nonverbal Reading Approach (NRA)	Not specified	Multiple-treatment design with drop-down baselines	Number of vocabulary words identified correctly	Not specified	Baseline, intervention across three conditions (teacher only, teacher + CAI, and CAI only)	Visual

Skills targeted: two studies have targeted receptive identification of pictures and vocabulary, one study has targeted matching words with the orthographic symbols, and one study targeted identification of words using the Nonverbal Reading Approach (NRA).

Participants

The number of participants in each study is between three and six. The average number of participants across the studies is five.

Materials

Hetzroni and Shalem (2005) identified eight words for participants with the help of teachers and parents. Bosseler and Massaro (2003) and Massaro and Bosseler (2006) identified 18 and 24 vocabulary items, which were equally distributed into three and four groups, respectively. These items were selected from 18 categories. Coleman-Martin, Heller, Cihak, and Irvine (2005) have not specified the material used in their study.

Design

The authors of the studies mentioned in the table have mainly used three out of six designs of SSRD, which include alternating treatments, multiple-baseline/multiple-probe design, and multiple-treatment design.

Data collected

It can be seen that across the studies, data collected is in terms of the number of correct responses given by the autistic children against the questions asked.

Duration of the study

Bosseler and Massaro (2003) have specified the entire duration as 33 days, whereas Massaro and Bosseler (2006) have only specified the duration of each session, which was conducted thrice a week. They have further mentioned that these sessions were performed until the child correctly identified all 24 vocabulary items in two consecutive posttests. The two studies (Coleman-Martin et al., 2005; Hetzroni & Shalem, 2005) have not specified the duration of the study.

Data collection instances

The data collection instances show that data was collected before (baseline or pre-training), during (intervention or training), and after (generalization, maintenance, or post-training) intervention.

Analysis

The results from the SSRD can be interpreted using various methods of statistical analysis (Dugard, File, & Johnathan, 2001) as cited in Horner et al. (2005). However, the visual analysis is the traditional method to analyze data gathered during each phase of the design (Horner). It can be seen that the authors of all the studies have used visual analysis rather than statistical analysis to examine the effectiveness of the intervention based on the data collected across the studies.

Outcome

The results of all studies have shown that learning among participants improved from the baseline to the intervention and generalization and maintenance.

Section 5 introduced the general aim of this experimental evaluation. A refined version of the aim is presented here based on the studies presented in Table 6. The aim of this experimental evaluation is to examine the effectiveness of the SG prototype entitled "vocab builder" (section 3) using the ABA design of SSRD in improving the receptive identification of vocabulary items among children with autism before (during baseline) and after (during intervention and maintenance following withdrawal of intervention) using the prototype. The improvement in receptive identification of vocabulary items is measured in terms of the number of correct responses given and the number of attempts made to identify the correct responses.

12. Method

Table 6 shows that each study has used a different design of SSRD; in this research, the ABA (basic withdrawal) design of SSRD is used. The purpose of using this design is to examine the effectiveness of using the SG prototype to improve learning of vocabulary among children with ASD by observing their behavior over a period of time, i.e., before (during the baseline phase), during the intervention, and maintenance tests at the end of week one and week two following the withdrawal of intervention.

12.1. Participants

For this research, a total of five children with autism were recruited to be a part of this evaluation. This number is marginally higher than the number of participants recommended for SSRD, as presented in Table 6. However, this number is the same as the average number of participants used in the studies related to the language comprehension of children with ASD (Table 6). The following criteria were used to recruit participants for this study.

(1) Child is diagnosed with ASD (from "requiring support" to "requiring very substantial support")
(2) Child has difficulty in learning different categories of vocabulary, for instance, fruits, vegetables, numbers, alphabets, etc.
(3) Child has a basic working knowledge of computers
(4) Child is within an age group of 6–10 years old

The participants were recruited from CADS Enhancement Centre, which is located in Kuala Lumpur, Malaysia, and provides intervention of different skills to children with ASD. Parental consent was sought before participants were involved in this experimental work. Brief information about each participant is presented in Table 7.

12.2. Study protocol

The principal of the center was briefed about the research and the experimental evaluation of the prototype and was invited for their students to take part in this study. The principal

Table 7. Information about the participants.

Participant#	Gender	Age	Diagnosis
Participant 1	Male	9	Requiring support & ADHD
Participant 2	Male	8	Requiring support
Participant 3	Male	10	Requiring support
Participant 4	Male	6	Requiring support
Participant 5	Male	7	Requiring support & ADHD

accepted the invitation and nominated one female teacher working with children with ASD as a focal person to provide all the support needed throughout this experimental evaluation. The teacher was informed that: 1) the identity of children participating in the study would remain anonymous, and they would be referred to as "Participant 1," "Participant 2," and so on; 2) pictures of participant's interaction with the SG will be taken, but no face of any participant would be revealed in any of the reports written based on this study.

The focal person was asked to identify five children who face difficulty in learning one or more categories of vocabulary mentioned in Table 3.

From Table 6, it can be seen that Massaro and Bosseler (2006) continued evaluation until participants met the criteria set by the authors. Bosseler and Massaro's (2003) is the only study that has mentioned the duration of the research, i.e., 33 days; the researchers used this duration to provide intervention of 18 vocabulary items from 28 categories. In this study, however, a total of 24 working days were reserved for the evaluation based on the availability of participants. This duration is solely used for the intervention of the vocabulary item bird. A slot of 2 h (20 min per participant in which they were given 2 min break toward the end of 10 min) per day across five weeks was reserved for the evaluation. Table 8 provides a summary of all the tasks that were performed during the five-week sessions. The total number of days per session and the time duration for which each participant would be allowed to use SG per day were fixed after consultation with the focal person. The number of days for intervention is kept considering the number of items to be learned by each participant. It would take an average of 10 days if two items (10 min per item) are learned each day. The focal person highlighted that some participants may need a recap on some of the vocabulary items and some participants may not be present on any day; therefore, additional days were added for the intervention.

12.3. Instruments used

The SG was developed and ran on an HP Envy 17 Notebook PC that has Intel(R) Core(TM) i7-4702MQ CPU @ 2.20GHz with 8 GB memory and NVIDIA GeForce GT 750M and Intel (R) HD Graphics 4600 graphics running Microsoft Windows 8.1 64-bit operating system. The environment setup for the intervention is shown in Figure 11. It shows that the researcher is sitting next to the participant for assistance if needed and when the participant is not sure what to do.

Table 8. Summary of the tasks performed during the three-week sessions.

Week	Days	Session	Participants	Tasks
0	2	Pre-session	All	Introduction to serious game "vocab builder" and how to use it
1	3	Baseline	All	Initial assessment (section 5.7.1)
1–4	15	Intervention	All	Learn vocabulary (section 5.7.2)
5	2	Maintenance 1	All	Test at the end of week one after intervention was over (section 5.7.3)
5	2	Maintenance 2	All	Test at the end of week two after intervention was over (section 5.7.3)

12.4. Material used

The prototype provides support in learning 11 different categories of vocabulary items as presented in Table 3. The focal person was asked to identify the categories in which recruited participants face difficulties so that intervention of specific vocabulary items can be provided through the prototype. The person identified birds, fruits, people, and colors and shapes as more crucial than others. Therefore, in this study, all 20 vocabulary items in the category of birds were selected with the consent of the focal person; these items represent 9.56% of all the items supported by the prototype. From Table 6, Bosseler and Massaro (2003) used 18 (out of 169, i.e., 9%) of all the items, which are almost the same number of items used in this study. Similarly, Massaro and Bosseler (2006) used 24 (out of 169, i.e., 12%) of all the items, which are marginally higher than the items used in this study. Both studies have used items from a collection of 28 categories; this shows that there is less than 1% chance to select an item from each of the 28 categories. This makes it difficult to make an overall understanding of a participant's knowledge in terms of each category. Therefore, in this study, seven different vocabulary items were chosen from the birds to gather performance during baseline and maintenance; these items were carefully identified with the help of the focal person to ensure that overall understanding of birds can be measured through these items. The purpose of using seven (out of 20, i.e., 30% of birds) was to get an understanding of participants' knowledge

Figure 11. Setup of environment for the intervention.

in distinguishing birds before and after using prototype. All 20 vocabulary items were used in the intervention in comparison to the number of items used during baseline and maintenance.

12.5. Procedure

A training session was conducted with the participants before beginning the formal investigation of the SG. The participants were trained to sit in a chair, use a computer, listen to verbal instructions given by the computer, and respond on screen by using the mouse.

The procedure used to collect data at three different stages, i.e., baseline, intervention, and maintenance, is described below.

12.6. Baseline

An initial assessment in the form of six baseline probe sessions was conducted before participants started using the SG to learn vocabulary. This assessment was conducted to measure the current level of knowledge in the identified categories of vocabularies. Three sessions were conducted on the first day with a break of 2 min in between each session, while the remaining three sessions were conducted on the second day. For each item, a question is asked to identify the correct image of an item by showing three images, including the item itself and two distractors from the same category as the item. The sequence in which seven vocabulary items need to be identified by the player, along with the position of the correct item and two distractors, is all randomized by the game so that the player cannot predict the question sequence of the item and the position of the correct item. This technique is also followed by intervention and maintenance.

12.7. Intervention

During the intervention, each participant went through all the vocabulary items present in the identified category. Two sessions of 10 min each were conducted with every player and a break of 2 min between sessions was given to the participants to relax. During each day, if a player was found not focused in the first session, then the second session was not conducted on that day. Each session corresponds to the learning of one vocabulary item from the identified category. For each vocabulary item, once the participants saw and memorized all the images associated with the item, they attempted a practice test where they were asked to identify the current image of the item they had just learned. Each player needs to identify at least three correct vocabulary items before they can go back to either learn the same item again or choose and learn another item.

Once learning of all the birds was completed by an individual player, he or she was given an access to play an activity game for two days. Each day, the player played an activity game once for the duration of 3 min; this duration was set after consultation with the focal person keeping in mind that children with ASD easily get distracted and bored. The activity game was specifically configured with the help of the focal person for the players as follows:

(1) Test items: birds
(2) Distractor items: animals and fruits
(3) Total number of items (test items and distractor items): four (three test items and one distractor)
(4) Background image of the game and its opacity: clouds with opacity of 0.8
(5) Animated items that can move around the screen: snail and snake
(6) Type of game (easy, medium, and hard): easy

While the practice test was conducted at each vocabulary item level, these activity games involve all the vocabulary items in the category. The purpose is to measure the extent to which a player can clearly identify the correct vocabulary items in the presence of distractors.

12.8. Maintenance

Three probe sessions of maintenance were conducted at the end of week one and week two following the withdrawal of intervention to measure the remembrance of vocabulary items learned during the intervention. During each week, two sessions were conducted on the first day, while the last session was conducted on the second day. If two sessions were conducted on the same day, then a break of 2 min was given in between each test. The same set of items from baseline was used in the maintenance; however, the images used for these items were those used in baseline or intervention.

12.9. Data measurement and analysis

This subsection describes the measurements performed during this evaluation. The approach used to measure the correct responses and the attempts made to identify correct responses during each session are the same in all three phases (baseline, intervention, and maintenance).

12.10. Measure the frequency of correct responses

The following variables are used during each session:

(1) Number of correct responses: the value of this variable is reset to 0 at the start of the session.
(2) Total number of responses: the value of this variable is reset to 0 at the start of the session.
(3) Response: For each question, if the player selected a correct response for the question, then the value of 1 is assigned to the variable and the value of a variable number of correct responses is incremented by 1; otherwise, the value of 0 is assigned to the variable. The value of the variable total number of responses is also incremented by 1.

The following formula is used to calculate the percentage of correct responses at the end of each session:

$$Percentage\ of\ correct\ responses = \frac{(number\ of\ correct\ responses * 100)}{total\ number\ of\ responses}$$

12.11. Measure the frequency of attempts made to identify the correct response

The variables used to calculate the number of attempts are as follows:

(1) Number of attempts: this variable is initialized to 1 when a new question is shown to the player.
(2) Sum of attempts: this variable is initialized to 0 at the start of the session.
(3) Total number of questions: this variable is initialized to 0 at the start of the session.
(4) Attempt: the value of this variable is initialized to 0 when a new question is shown to the player. If the player selected an incorrect response for the question, then the value of a number of attempts variable is incremented by 1. If the player selected a correct response for the question, then the value of the number of attempts variable is added to the sum of attempts variable and the value of the total number of questions is incremented by 1.

The following formula is used to calculate the value of a number of attempts at the end of each session:

$$number\ of\ attempts = \frac{sum\ of\ attempts}{total\ number\ of\ questions}$$

12.12. Examine the effectiveness of the SG prototype

Visual analysis was also used to examine the effectiveness of the prototype based on the two parameters: 1) number of correct responses and 2) number of attempts. In the visual analysis, each graph of the participant is examined one after another to assess the change that occurs in the behavior from the baseline to the intervention and maintenance.

13. Results

The percentage of correct responses and the number of attempts made to identify the correct items during each session of all the participants are shown in Figure 12–16 in sections 5.9.1–5.9.5. The upper part of each figure shows the percentage of correct responses, while the lower part shows the number of attempts made to identify the correct responses. Each figure is divided into four parts through vertical dash lines: the first part contains performance during baseline. The performance of intervention is shown in the second phase, while the performance in the maintenance, i.e., at the end of weeks 1 and 2 following the withdrawal of intervention, is shown in the third and fourth parts, respectively.

13.1. Performance of participant 1

Figure 12 displays the percentage of correct responses and the number of attempts made to identify the correct items by Participant 1. The following subsections present the results in terms of: a) number of correct responses and b) number of attempts made to identify the correct responses.

Correct responses

The mean performance of the participant during baseline was 72%. The use of SG during the intervention to learn different birds improved the performance of participant 1 to 97%. The participant had remembered 100% vocabulary items during maintenance at the end of weeks 1 and 2 following the withdrawal of intervention.

The results of the post-intervention activity game played on the first and second days show that participant 1 correctly identified 55 out of 55 (i.e., 100%) and 43 out of 44 (i.e., 97.72%) birds in each of the 3-min activity games.

Number of attempts

The mean number of attempts made to identify the correct bird during baseline was almost one and a half, i.e., 1.4, which

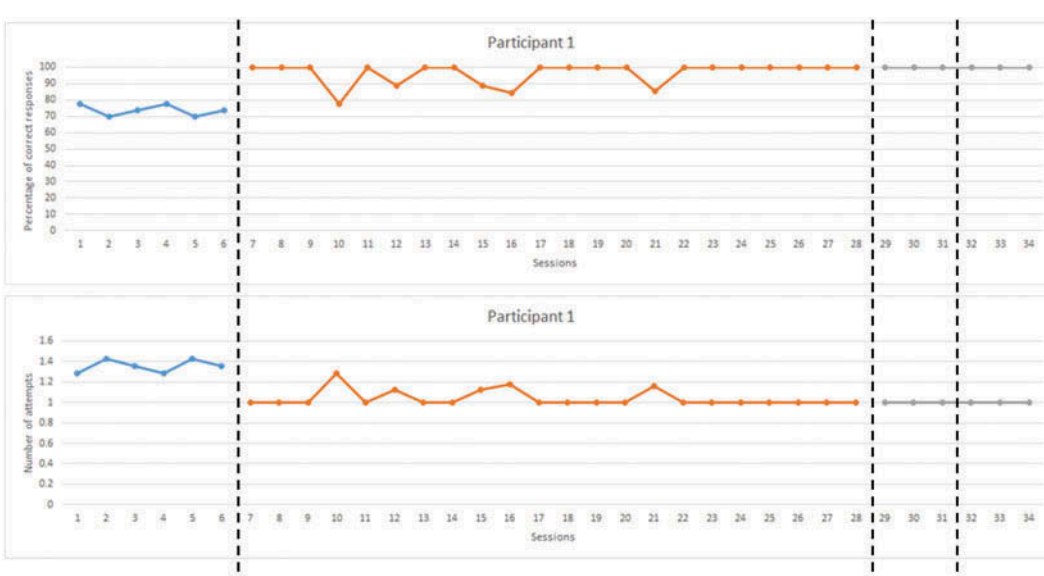

Figure 12. Performance of participant 1.

becomes lower and improved to 1 after learning through SGs and it remained the same during maintenance at the end of weeks 1 and 2.

13.2. Performance of participant 2

Figure 13 displays the percentage of correct responses and the number of attempts made to identify the correct items by Participant 2. The following subsections present the results in terms of: a) number of correct responses and b) number of attempts made to identify the correct responses.

Correct responses

The mean performance of the participant during baseline was 53%. The use of SG during intervention brought a lot of improvement in the performance of participant 2 and the performance increased to 94%. The participant had remembered 92% of the vocabulary items during maintenance at the end of weeks 1 and 2 following the withdrawal of intervention.

The results of the post-intervention activity game played on the first and second days show that participant 2 correctly identified 50 out of 53 (i.e., 94.34%) and 69 out of 70 (i.e., 98.57%) birds in each of the 3-min activity games.

Number of attempts

The mean number of attempts made to identify the correct bird during baseline was 1.8, which becomes lower and improved to 1.1 after learning through SGs, and it remained the same at 1.1 during maintenance at the end of weeks 1 and 2.

13.3. Performance of participant 3

Figure 14 displays the percentage of correct responses and the number of attempts made to identify the correct items by participant 3. The following subsections present the results in terms of: a) number of correct responses and b) number of attempts made to identify the correct responses.

Correct responses

The mean performance of the participant during baseline was 59%. The use of SG during intervention brought an improvement in the performance of participant 3 and the performance increased to 81%. The participant had remembered 88% of the vocabulary items during maintenance at the end of weeks 1 and 2 following the withdrawal of intervention.

The results of the post-intervention activity game played on the first and second days show that participant 3 correctly identified 19 out of 21 (i.e., 90.48%) and 24 out of 26 (i.e., 92.31%) birds in each of the 3-min activity games.

Number of attempts

The mean number of attempts made to identify the correct bird during baseline was 1.6, which becomes lower and improved to 1.3 after learning through SGs, and it remained almost the same, i.e., 1.2, during maintenance at the end of weeks 1 and 2.

13.4. Performance of participant 4

Figure 15 displays the percentage of correct responses and the number of attempts made to identify the correct items by participant 4. The following subsections present the results in terms of: a) number of correct responses and b) number of attempts made to identify the correct responses.

Correct responses

The mean performance of the participant during baseline was 42%, which shows that participant 4 was facing difficulty in identifying the correct items in the start. The use of SGs during intervention brought a lot of improvement in the performance of participant 4 and the performance increased to 96%. The participant had remembered 100% vocabulary items during maintenance at the end of weeks 1 and 2 following the withdrawal of intervention.

Figure 13. Performance of participant 2.

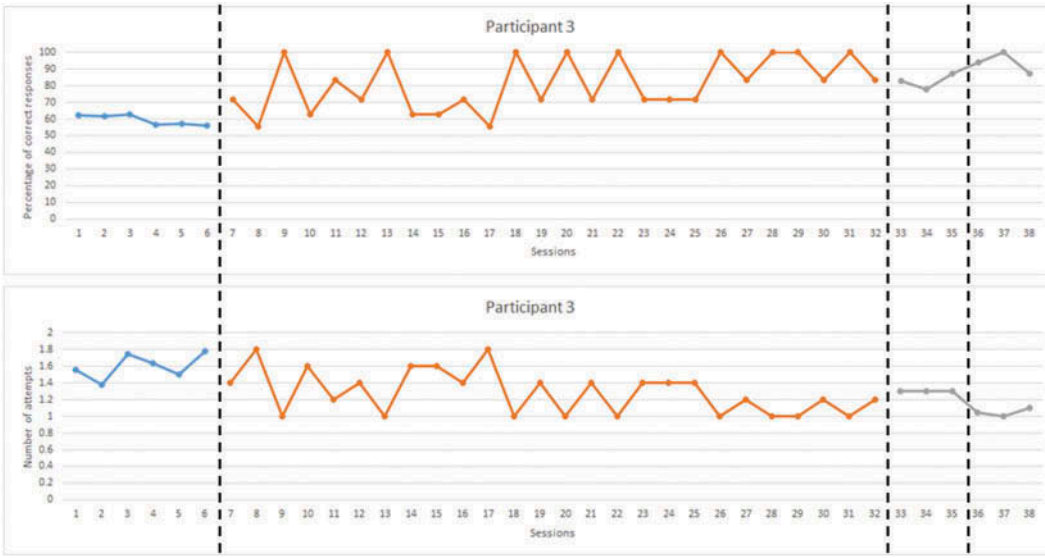

Figure 14. Performance of participant 3.

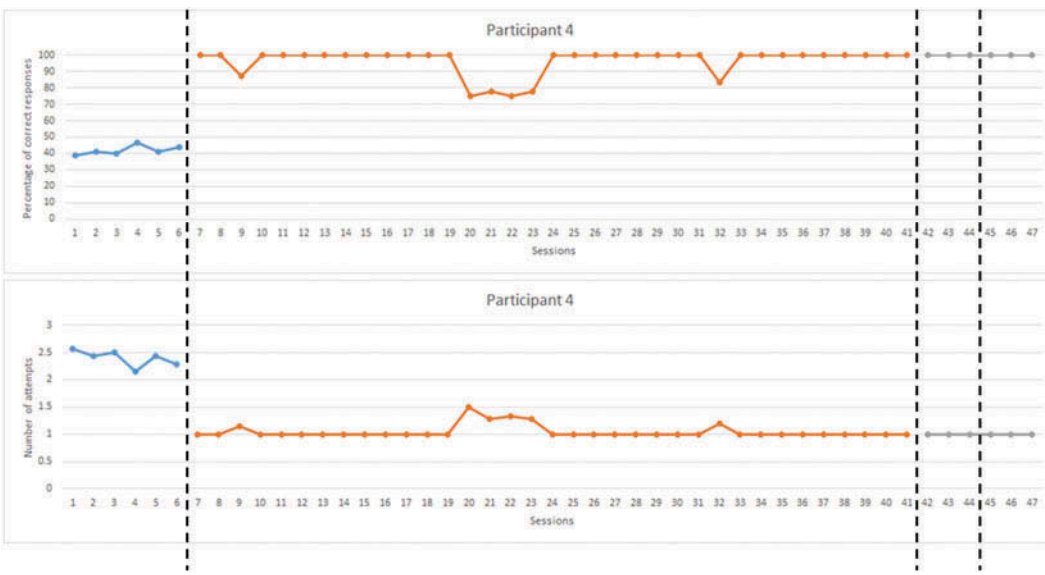

Figure 15. Performance of participant 4.

The results of the post-intervention activity game played on the first and second days show that participant 4 correctly identified 46 out of 48 (i.e., 95.83%) and 59 out of 61 (i.e., 96.72%) birds in each of the 3-min activity games.

Number of attempts

The mean number of attempts made to identify the correct bird during baseline was 2.4, which was lowered and improved to 1.1 after learning through SGs, and it remained almost the same, i.e.,1, during maintenance at the end of weeks 1 and 2.

13.5. Performance of participant 5

Figure 16 displays the percentage of correct responses and the number of attempts made to identify the correct items by participant 5. The following subsections present the results in terms of: a) number of correct responses and b) number of attempts made to identify the correct responses.

Correct responses

The mean performance of the participant during baseline was 42%, which is an indication that participant 5 was struggling to identify the correct items in the start. The use of SGs during intervention brought a lot of improvement in the performance of participant 5 and the performance increased to 89%. The participant had remembered 88% of the vocabulary items during maintenance at the end of weeks 1 and 2 following the withdrawal of intervention.

The results of the post-intervention activity game played on the first and second days show that participant 5 correctly identified 45 out of 45 (i.e., 97.78%) and 45 out of 49 (i.e., 91.84%) birds in each of the 3-min activity games.

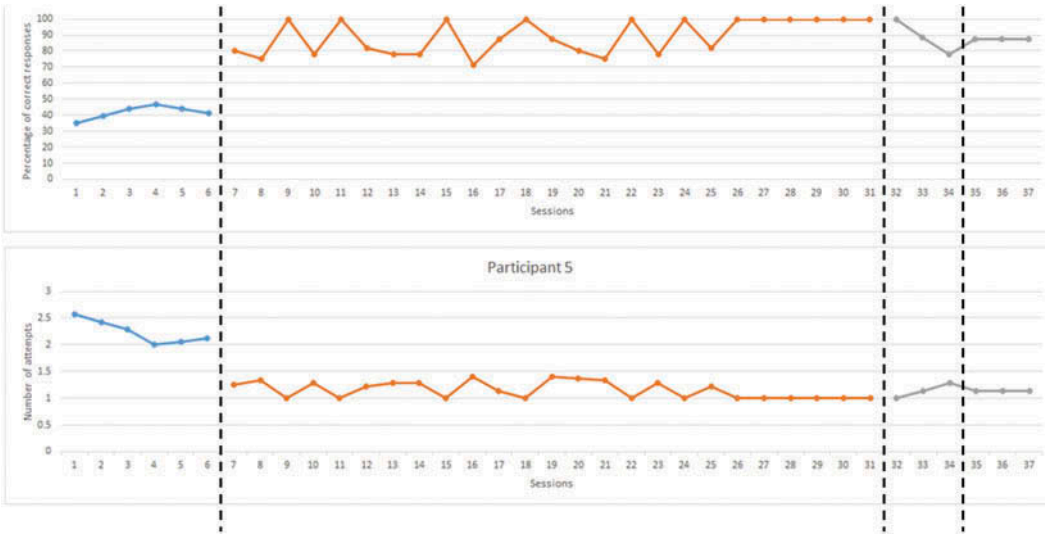

Figure 16. Performance of participant 5.

Number of attempts

The mean number of attempts made to identify the correct bird during baseline was 2.25, which was lowered and improved to 1.1 after learning through SGs, and it remained the same during maintenance at the end of weeks 1 and 2.

13.6. Effectiveness

Correct responses

The visual analyses of the correct responses across the participants in Figures 12–16 show that the number of correct answers improved from the average of 53.97% during baseline to 92.57% during intervention, i.e., when they were playing SGs for the learning. The participants also had remembered 93.73% of the vocabulary items during maintenance at the end of weeks 1 and 2 following withdrawal of intervention

Number of attempts

The visual analyses of the number of attempts made to identify the correct responses across the participants in Figure 12–16 show that these attempts were reduced from a mean of 1.9 during baseline to 1.11 during intervention and remained the same during maintenance

13.7. Overall performance

The result suggests that players were able to learn distinguishing birds through the prototype and identify specific birds when they are asked through a written or verbal question. Their learning of birds improved after using the prototype and they retained the names of birds they learned after the first and second weeks once the intervention was withdrawn.

The results also suggest that as the learning of players improved through the intervention, their number of attempts made to identify the correct responses was also reduced from baseline to intervention and remained the same in maintenance.

14. Conclusion

The analysis of the results from the experimental evaluation of the SG prototype shows that the number of correct responses to receptively identify the vocabulary items among children with ASD improved from the baseline to intervention and maintained at the end of weeks 1 and 2 following the withdrawal of intervention. The number of attempts made to identify correct responses lowered as the number of correct responses improved. This shows that prototype was effective in facilitating children with ASD to learn vocabulary.

The design of the SG prototype, which is based on the SGDF presented in this research, has:

(1) four specific components (autism behaviors and symptoms, instruction method, instructional strategies, and modalities) from the ASD perspective that can contribute to the learning of vocabulary among children with ASD. These components are also equally important as of any other components used in the SGs for children with ASD; however, these components have not been emphasized in any existing SGDFs for children with ASD, typical children, or typical users. It is important to highlight how the components from the ASD perspectives contributed in the learning of children with ASD. The specific instruction method called DTT used in the design provided rigorous learning and practice of each vocabulary item in the category of bird and improved the skill of receptive identification of items. The use of strategies like multimedia instruction provided audio and visual learning, while explicit instruction provided details and multiple exposures of all the birds. The most commonly used modalities were incorporated in the design so that it is easier for the child to interact with the environment of the SG. This indicates the applicability of the framework to design SGs that can provide an effective learning environment for children with ASD to learn vocabulary.
(2) components from existing frameworks for typical children and typical users that can contribute to various

aspects of the SG. These include from identifying desirable capabilities and associated learning outcomes to the identification of appropriate instructional contents and creating a user profile that can incorporate all the information about the player and its progress and achievements in the game. It also includes carefully selecting the specific GBLAs to be incorporated in the SG where each GBLA is supported by one of the learning theories or theory of psychology.

Altogether 14 out of 20 components from the framework were used in the design of the SG. As mentioned earlier, the SG prototype presented in this research article was produced by one of the experts as a part of applicability evaluation of the SGDF. This shows that the expert was able to read through all the components and their descriptions, understand the relationship between the components, and brainstorm about the components that would be required to design SGs for children with ASD. This also shows that the framework can be used by any individuals including game designers and researchers working with children with ASD among others to produce designs for them.

For the future work, the researchers can:

(1) consider different categories of vocabulary items or vocabulary items that have not been covered in the categories used in the prototype;
(2) incorporate more modalities in the SG and determine its impact on the learning of vocabulary;
(3) develop a virtual reality-based environment where children are exposed to real-world scenarios. For instance, if they are to be taught different food items, they are shown a scenario of one grocery store with different food items placed in different aisles of shelves and they are asked to visit all the aisles in the store and explore the items placed in the shelves.

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Appendix A Components of serious game design framework

Input

Following are the components and their description of components in this phase.

A1.1. Autism behaviors and symptoms

Each child with ASD possesses a different set of behaviors. Before the game design process begins, it is important to identify the behaviors that need to be targeted in the serious game. The selection of these behaviors will help in the selection and creation of appropriate instructional contents and learning activities. Table A1 presents all the behaviors related to ASD, which have been classified into language comprehension, verbal communication, nonverbal communication, and general behaviors.

Table A1. Categorization of behaviors associated with social-communication skills.

Language comprehension	Verbal communication	Nonverbal communication	General behaviors
1. Content and grammar may be delayed 2. Rigid understanding of words 3. Difficulty with the concept that objects can have more than one name 4. Difficulty to identify nouns 5. Reverse pronouns (e.g., says "you" instead of "I") 6. Inappropriate use of verbs 7. Inappropriate use of adjectives 8. Struggle to receptively or expressively label places, people, objects 9. Language may be slow to develop	1. Speech may be delayed, or there may be no speech at all 2. Might not know how to start, sustain, or end verbal conversations 3. Frequently use echolalia (repeating words of others) 4. Show unusual tone of voice (e.g., Monotone, robotic, or high pitch) 5. No consistent response to name 6. Unable to express wants and needs 7. Talk in a flat, robot-like, or singsong voice 8. Less likely to make comments or ask questions 9. Less likely to make bids for social attention	1. No effort to use nonverbal communication 2. Use fewer gestures 3. Difficult to follow directions 4. Appear not to hear at times 5. Unable to point or wave 6. No accurate interpretation of puns, sarcasm, idioms, etc. 7. Do not pretend in play 8. Do not understand jokes, sarcasms, or teasings	1. No response in normal teaching methods 2. Less likely to share experiences 3. Give unrelated answers to questions

A1.2. Desirable capabilities

Capability refers to the development of one or more of three types of skills, namely cognitive, psychomotor, and affective skills that players of the game can develop as a part of playing the game (Yusoff, 2010). These skills are based on the cognitive, psychomotor, and affective domains.

Cognitive skills refer to the development of thinking ability in the players, which they use to solve problems of varying degrees, from easy to difficult, either as a part of learning activities or as evaluations following learning of contents through the game (Bloom, 1971). The categories of cognitive skills have been drawn from Bloom's taxonomy, which includes recall, understand, apply, analysis, synthesis, and evaluation.

Psychomotor skills are related to the development of motor skills, physical movements, and coordination of body parts so that they can manually perform all the tasks (E. Simpson, 1971). The psychomotor skills have not been categorized by Bloom and his colleagues; however, there are other researchers who have categorized the skills and created their own psychomotor taxonomies. These include Simpson's psychomotor domains (E. Simpson, 1971), Harrow's psychomotor domains (Harrow, 1972), and Dave's Psychomotor domain (Dave, 1970). The categorizations by Simpson and Harrow are useful for the development of this skill from children to adults. Table A2 shows how the psychomotor skills have been categorized by different researchers.

Table A2. Categorization of psychomotor skills by various researchers.

Sources	Categorization
(Simpson, 1971)	1) perception (awareness) 2) set 3) guided response 4) mechanism (basic proficiency) 5) complex overt response (expert) 6) adaptation 7) origination
(Harrow, 1972)	1) reflex movement 2) fundamental movements 3) perceptual abilities 4) physical abilities (fitness) 5) skilled movements 6) no discursive communication
(Dave, 1970)	1) imitation 2) manipulation 3) precision 4) articulation 5) naturalization

Affective skills are all about learning how to deal with things emotionally, which include feelings, values, appreciation, enthusiasm, motivation, and attitude (Krathwohl, Bloom, & Masia, 1967). The taxonomy created by these authors for the affective domain includes five categories, namely receiving, responding, valuing, organization, and value complex.

A1.3. Intended learning outcomes

The intended learning outcomes are the overall goals that need to be achieved by playing a serious game (Garris et al., 2002; Yusoff, 2010). It also refers to the fact about what players should be able to do once they have completed playing the game (Kennedy, Hyland, & Ryan, 2006). Each outcome of the serious game is created based on the combination of specific capability to be achieved in the user, as well as specific instructional contents. Each learning activity to be incorporated into the games should have at least one learning outcome to be achieved by the child (player of the game). These outcomes are decomposed into all the activities of the games based on the overall instructional contents to be learnt by players. This way, players know in advance what needs to be accomplished and what to expect from the material to be learnt within a specific activity. According to the pedagogical guidelines based on Gagne's nine events of instruction (Gagne, 1985), these learning outcomes should be communicated to the user. The outcomes are typically shown at the introduction of the serious game and also available through the help menu of the serious game.

A1.4. Instructional contents

The instructional contents refer to the specific contents or subject matter that players can learn through the serious game. The categories of instructional contents and specific contents supported by the serious game could be an exhaustive list. According to Gilbert & Gale (2008), there are four different types in which contents can be classified, namely, facts, concepts, procedures, and principles. Each of these types is briefly described below.

Fact: it is a statement that is true or real; it could also refer to something that can be shown to be real or true. It consists of two specifics "A" and "B," the so-called fact pair about something; these specifics could refer to definition, name, or number. For instance, one liter of milk weighs one kilogram or Malaysia is a federation consisting of 13 states (Negeri) and three federal territories (Wilayah Persekutuan).

Concept: means basic understanding about something that is necessary to make sense out of it. This type of learning is typically based on values, ideas, symbols, events, or things that are associated with it. A concept has name "X," a superordinate class "Y" of which X is a member and a list of attributes "An" and values "Vn" where "Vi" corresponds to "Ai." These attributes serve a purpose to classify or sort objects of different types. The use of the phrase "is a" distinguishes a concept from that fact. For instance, "an apple" is a "fruit" and the list of attributes could be "color," "weight," "country of origin," and "taste" among others.

Procedure: refers to learning how to step-by-step perform a sequence of actions to accomplish a task. A procedure typically has a name "X" and it is used in some situation "S" to achieve goal "G" using a set of

steps "E" by using some tool "T." For example, a person needs to follow these steps to eat a pomegranate: first, remove its crown with a sharp knife; then score the pomegranate with cuts if you plan to divide it into quarters; then soak it in cold water as it helps in easy collection of seeds. While the pomegranate is still in the water, slowly divide the pomegranate into quarters; then to separate the seeds, run your fingers through each quarter slice.

Principles: refer to a proposition that contain a name "X," are applied in a situation "S," and involve the specification of cause–effect relationships (CERs) between objects or events "E."

Process

The components that facilitate in designing learning activities are described as follows.

A2.1. Reflection

Serious games typically consist of a set of interrelated learning activities where each activity builds on top of previous activity in terms of the difficulty level of an activity. Reflection is about providing an opportunity to the player to think about reasons for undertaking the current learning activity (Yusoff, 2010). This thinking process is based on all the learning activities that the player has performed since it started playing the serious game. This allows the player to understand and identify the skills and approaches used to solve similar problems in the previous activities and identify approaches and skills needed to solve problems in the current activity.

This process can be facilitated by providing information to the players toward the end of each activity about: 1) the current activity that was undertaken, 2) their achievement in the activity, and 3) the objectives accomplished through the activity. This information could be useful for the player, as otherwise they may forget the actual objective behind the current activity. Garris et al. (2002) have also recommended a similar set of information to be provided to the player. The set of information includes: 1) description of the current activity undertaken, 2) purpose of undertaking the current activity, 3) all the errors made throughout the activity, and 4) suggestions on how to correct those errors. According to Simon (1979), Kolb's problem-solving cycle can also be used to support player's learning based on the reflection. This cycle includes four steps that be incorporated in the learning activity; these steps include doing, reflecting, understanding, and applying.

A2.2. Instruction method

Instruction methods for ASD are the evidence-based methods that have proven effective for some individuals on the spectrum (National Autism Center [NAC], 2009). The report by NAC states that there is no universal method as such that has proven to be the most effective method for all the children on the spectrum. The use of instruction methods in the intervention for children with ASD highly depends on the skills targeted in the intervention. The skill targeted throughout this research is the vocabulary of children with ASD and there are five different instruction methods that can be used to provide intervention of vocabulary in particular or language in general. These methods include DTT, treatment, and education of autistic and related communication handicapped children (TEACCH), incidental teaching, pivotal response training (PRT), and verbal behavior therapy (VRT).

A2.3. Instructional strategies

Strategies are the techniques that facilitate instructors to get the attention of children, stimulate their interest toward the learning of contents, keep them engaged in the learning process, develop their thinking skills, and increase interaction so that they remain focused on the learning (Khowaja & Salim, 2013). The strategies can also be used to connect the idea to the real world, which can provide an opportunity to children with ASD to generalize the contents of one situation to another. These strategies are used to support instruction methods for ASD.

A2.4. Learning activities

Learning activities are the set of activities designed to keep players actively involved when they are playing the games. The effective design of these activities ensures that players stay engaged and immersed without getting bored. Another important aspect is to ensure that the learning material used is appropriate and challenging for players who are seeking improvement in their competency slightly above their current level of competency (Gee, 2007). These activities make use of the material from the instructional contents to be used throughout the game.

A2.5. Game genre

Game genre is a term used to categorize games based on the interaction between the players' so-called gameplay rather than differences in visual or narratives (Yusoff, 2010). It is defined by a set of gameplay challenges. These genres range from action-adventure to a combination of action-adventure, role-playing, simulations, strategy, and sports, among others.

A2.6. Game mechanics

Game mechanics are a set of actions, behaviors, and control mechanisms given to a player within the context of a game (Hunicke, LeBlanc, & Zubek, 2004; Schreiber, 2009; Winn, 2008). These mechanics are used to define a rule-based system for the game environment by specifically stating all the objects that would be available within the game environment, how each object would behave, and how the user can interact with them in the game world.

A2.7. Game dynamics

Game dynamics refer to the change that occurs in the gameplay when the game mechanics are activated within the environment of the serious game (Schreiber, 2009; Winn, 2008). This provides more fun, enjoyment, and engagement in the games for the player. Assume the user is playing a time-based game in which the user needs to collect specific fruits falling from trees (placed at a different horizontal location with varying heights) in the basket before they touch the ground. These fruits are mixed with other fruits that act as distractors. The user is penalized for catching one of the distractors. Both the actual and distractor fruits fall at the same constant speed until 30 s remains, after which the gameplay changes as the speed is increased by twice and the penalty factor is also doubled.

A2.8. Game story

The overall story of the game consists of one or more of the three components, i.e., storytelling, narratives, and the characters involved in the game. These three components of the game story are described in the following subsections.

A2.8.1. Storytelling

The use of the game as a medium for storytelling has largely increased in the past decades (Sinclair, 2014). This has become possible because of the improvement in technology due to which game designers are able to bring characters in the game environment. Sinclair states that "Games have done more work to pioneer best practices in interactive and participatory storytelling than any medium since the ancient theatre traditions."

The storytelling in a serious game can be in two different forms: 1) designer's story; and 2) player's story (Rouse, 2010), as cited in Winn (2008). A designer's story highly depends on the integration of various aspects related to the serious game, i.e., the context in which a serious game is to be played, instructional contents to be learned by the user, learning activities to be created, and how characters and narratives are synchronized with each other and used in different activities. The resultant story is incorporated into a serious game. The designer tends to describe the situation to others exactly the way they perceive it (Rouse). The storytelling that the user experiences while playing the serious game is based on the designer's version of the story, as well as all the interactions and the selections made by the user itself throughout the serious game. The overall resultant experience becomes the player's own version of the story, and it is unlikely that the stories of two players would be the same. The use of the designer's story highly depends on the game genre of the serious game, i.e., their story is strongly used in the adventure or role-playing games, whereas in game genres like logic or board games, it is possible that the designer's story is little-used or not used at all. However, each serious game does include a player's version of the story, which at the minimum includes all the activities played and the challenges performed throughout the game. It is at the discretion of the game designer to decide which type of story should be experienced by the player of a serious game.

A2.8.2. Narrative

A narrative refers to one of the story events presented to the player within the game world (Adams, 2010). It consists of a noninteractive presentational material in the form of either text only or a combination of text, audio, and animation shown to the user as a part of the narration. As Adams described, the purpose of the narrative is to present an event that cannot be controlled or interacted by the user. The narrative is typically used to establish the setting of an environment in which a player would provide initial motivation toward the game, but often, it is not the main focus of gameplay. It is also shown when the user moves from one activity or type of content to another, toward the end of the level when the user has

accomplished the goals or toward the end of the game as well. Narrative allows designers to keep players engaged in the game through the way story is told (Sinclair, 2014). Although the players already know that the outcome of the game is already predefined, it keeps them engaged because of the uncertainty of events happening through the game and the way the story progresses.

A2.8.3. Characters

Aarseth (2012) has categorized characters into three kinds based on their depth/shallowness and malleability/potential for players to control characters. These categories include: 1) bots (Techopedia, 2016), which are also referred to as robot and sometimes interchangeably used with the term non-player character (NPC); 2) shallow characters, which are given names and also have a personal appearance in the game but are given limited personality; and finally 3) deep characters, which are fully controlled by the player itself.

The players of real-time games excluding strategy games typically control one or more number of characters within the game environment and play against the set of characters either controlled by other players as typically happens in a multiplayer environment or controlled by the computer itself (Kienzle, Denault, & Vangheluwe, 2007). These authors have referred to these computer-controlled characters as NPCs; the behavior of these characters is controlled by a piece of code specially written in artificial intelligence (AI) that their actions, reactions, and appearance are as natural as possible. As a part of their research, they have presented the model-based approach to control the behavior of these NPCs.

Thus, the role of each character used within a serious game may vary from one another. They may be used to provide support to the user whenever they are stuck within the game environment and not sure what to do next. They can be part of the user's team in a strategy-based game to support the player in achieving her goals or they can even be on the opposite team as your enemy. These characters can also play the role of some random character who is freely moving around within the game environment and may or may not have an ability to interact with the character when both are next to each other.

A2.9. Game-based learning attributes (GBLAs)

The learning through serious games needs to be validated against current teaching practices to ensure it is at the same level as that of traditional classroom learning (Garris et al., 2002; Yusoff, 2010). Therefore, it is important to incorporate certain aspects of current teaching practices in serious games that have proven to be successful in the classroom. These successful aspects are referred to as GBLAs. These GBLAs support learning and engagement throughout the game; this allows users to remain active and think critically while learning and playing games. Each attribute in GBLAs is based on different theories, namely behaviorism, cognitivism, constructivism, and theory of psychology. The following table shows a list of all the attributes and their classification into one of the four mentioned theories.

Table A3. Mapping of game attributes to the theories of learning and theory of psychology.

	Behaviorism	Cognitivism	Constructivist	Psychology
Attributes	Interaction (equipment) Interaction (interpersonal) Interaction (social) Language or communication Reward Practice and drill Feedback Rules/Goals Positive feelings Conflict Debriefing	Incremental learning Linearity Attention span Transfer of learnt skill Motivation	Scaffolding Learner control Progress Adaptation Location Surprise Help and support	Situated and authentic learning Accommodating the learner's style Assessment Fantasy Pieces or players Representation Sensory stimuli Safety Challenge Mystery Intensity

A2.10. Modalities

Modality acts as a channel of communication between the user and the computer (more precisely, the serious game in the context of this research) itself (Nigay & Coutaz, 1993). It can be used to convey some information to the serious game or acquire some information from the serious game in return. Games that make use of just one modality are known as unimodal, whereas games in which two modalities are used are known as bimodal. Finally, games that integrate more than two modalities are known as multimodal games (Oviatt, 2003). The combination of two or more modalities for input is known as multimodal fusion. Likewise, the partitioning of information into two or more communication modalities is also known as multimodal fission. Examples of input modalities include speech, gestures, gait, and facial expression among others, whereas text, graphics, animation, video model, virtual character, and force feedback are a few examples of output modalities.

Output

The components of output are described next.

A3.1. User profile and achievements

The game must allow users to set up their own profiles that contain their basic information. Once created, this profile will be updated regularly with all the accomplishments as users progress through various activities of the games. This way, users can also look back at all the achievements. It can also allow them to practice one of the previously learnt skills to improve their learning. Achievement in the games can be shown in the form of scores, a total number of resources gathered in the games, or the time taken to accomplish that task.

A3.2. Debriefing

The purpose of debriefing is to provide a review of all the activities in which users took part while playing the game (Garris et al., 2002). For each activity, this review may provide a description of the activity performed, their performance in the activity, highlighting all the incorrect responses as well as the correct responses that should have been selected. The use of debriefing or post-experimental analysis in the educational setting was first reported by Lederman (1984); it was first used in the simulation or game. Debriefing plays an important role in the serious games to improve the overall learning and experience of the user (Garris). Consideration needs to be given on how to display the information so that users do not feel overloaded and can easily concentrate on the corrective actions that need to be taken.

Appendix B Game design document

Part A: Information about the game

(1) Title of the game: Vocab builder
(2) Brief description of the game: This 2D game allows children with ASD to learn the various words from different categories through the audio-visual medium of instruction. The child is free to select one of the objects to learn and browse through all the images showing different views of an object; this facilitates the receptive skills. A child can also play different sounds and learn how to pronounce the name of an object and this can improve their expressive skills. Once a child has undergone the learning, it can take part in different activities to practice, learn, and improve its learning of different objects.
(3) Age group of children: 6–10 years
(4) Number of children who can play the game simultaneously: 1

Part B: Specification of the game's components

Game's components in input:

(1) Autism behaviors and symptoms: one common behavior, i.e., "struggle to receptively identify objects" among children with ASD, is targeted in the design of the serious game.
(2) Desirable capabilities: by the end of the game, it is expected that the child can identify the objects when they listen to the names of objects and recall the same objects over a period of time.
(3) Intended learning outcomes: the specific learning outcomes to be achieved in children with ASD by the end of this game are: 1) identify objects and 2) recall objects over the period of time.
(4) Instructional contents: the extensive learning of contents to be provided in the game includes fruits, birds, foods, numbers, and alphabets.

Game's components in the process:

(5) Reflection: this informs the child about the purpose of undergoing any activity, rectify errors made during activity, and provide necessary suggestions to overcome these problems in future.
(6) Instruction method: the instruction method to be used in the game is discrete trial training (DTT).
(7) Instructional strategies: the specific strategies of explicit instruction and multimedia instruction used in the game are described in the following table.

Table B1. Specific strategies.

Explicit instruction	Multimedia instruction
• Students are given definition or other attributes associated with the word to be learned • Repeated and multiple exposures of words lead to gain in vocabulary • The game contains images, stimulus materials, or instructional examples utilised in teacher-directed instruction • The game contains images, stimulus materials, or instructional examples found in the age-appropriate general education classroom	• Children are taught vocabulary by going beyond text and use other media, for instance, still and animated images, sound, and animated characters among others • Children who are unable to read yet will link text with spoken words. Moreover, autistic children do struggle with comprehending spoken words, especially those from TVs; and it may be beneficial to children who can read as they are able to see and hear the words • Instructions are provided in the form of written words, pictures where needed, and the verbal instructions are also simultaneously played to help them in understanding. These instructions are provided in small steps

(8) Game genre: it is an educational game that provides an interactive learning environment for different types of vocabulary items.
(9) Game mechanics and dynamics: the game mechanics and the game dynamics associated with this game are mentioned in Table B2.

Table B2. Game mechanics and game dynamics.

Game mechanics	Game type	Game dynamics
Aim and select	Easy	If the correct object is identified: • Name of the object is shown underneath the object • Name of the object is played for child to listen and memorize • Increase points by 10 If the incorrect object is identified: • Highlight all the correct objects present on the screen for 1 s to provide hint
	Medium	If the correct object is identified: • Name of the object is shown underneath the object • Name of the object is played for child to listen and memorize • Increase points by 10 If the incorrect object is identified: • Highlight one of the correct objects present on the screen randomly for 1 s to provide hint • Decrease points by 5
	Hard	If the correct object is identified: • Increase points by 10 If the incorrect object is identified: • No hint is provided like other levels • Decrease points by 5 • Decrease time by 5 s

(6) Game story: learning activity is composed of one scene.
(7) Storytelling: the story is focused on a child who faces difficulty in the receptive identification of an object when the name of the object is called. This game highly relies on the player's version of the storytelling, which means the player will create its own story by making all the choices from the start, i.e., selection of a category in the learning to the selection of a specific vocabulary item to learn as s/he progress in the game.
(8) Narrative: no narratives
(9) Characters: among the different types of characters, this game makes use of one helping character for the interaction with the player (children with ASD). Players can interact with the character to seek guidance when they are not sure what to do next. This character also monitors the activities in the game environment and communicates with players accordingly; for instance, if it does not recognize any activity (action performed through mouse) in 10 s, then it will auto popup in the middle of the screen to ask the players if everything is alright or are they stuck and looking for support.
(10) Game-based learning attributes (GBLAs): the specific GBLAs used in the game are presented in Table B3 along with their underlying theories.

Table B3. Specific game attributes and theories used in the game (Khowaja, 2017).

Behaviorism	Cognitivism	Constructivist	Psychology
(1) Reward (2) Practice and drill (3) Feedback (4) Rules/ Goals (5) Positive feelings	(1) Incremental learning (2) Attention span (3) Motivation	(1) Scaffolding (2) Learner control (3) Progress (4) Surprise (5) Help and support	(1) Situated and authentic learning (2) Assessment (3) Challenge

(11) Modalities: The modalities incorporated in the game include text, images, animated character, animation, and sound.

Game's components in output:

(12) User profile and achievements: each user profile includes name, gender, email address, date of birth, and all the achievements made in the game.